

Multivariable Calculus Quiz

Question 1 (5 points)

Find the gradient of the function $f(x,y) = x^2y + \sin(xy)$ at the point $(\pi/2, 1)$.

Question 2 (5 points)

Compute the directional derivative of $f(x,y) = 3x^2 - 2y^2$ at the point $(1,2)$ in the direction of the vector $v = [3, 4]^T$.

Question 3 (5 points)

Find the critical points of $f(x,y) = x^3 + y^3 - 3xy$ and classify them (local maximum, local minimum, or saddle point).

Question 4 (5 points)

Use Lagrange multipliers to find the maximum and minimum values of $f(x,y) = xy$ subject to the constraint $x^2 + y^2 = 8$.

Instructions: Show all your work. Partial credit will be given for correct setup even if the final answer is incorrect.
Circle your final answers.